A.Problem Statement Of the given assignment: To predict insurance charges

B.Basic info of the dataset: File name - insurance_pre; Rows and columns: 1338 rows × 6 columns.

C: Data preprocessing: Converted nominal data into ordinal data;
2 columns – SEX and SMOKER were nominal and hence those are converted into ordinal data with Onehot encoding.

D: R^2 Value for different Algorithms for the dataset of insurance_pre:

1.Multi Linear Regression : R_score = 0.7895

2.Support Vector Machine:

S.No	C Value (Hyperparameter)	R_Score				
		Linear	Poly	rbf	sigmoid	precomputed
1	1.0	-0.0101	-0.0757	-0.0834	-0.0754	-(Error due to matric requirement)
2	10	0.4625	0.0387	-0.0323	0.0393	”
3	100	0.6289	0.6180	0.3200	0.5276	”
4	500	0.7631	0.8264	0.6643	0.4446	”
5	1000	0.7649	0.8566	0.8102	0.2875	”
6	2000	0.7440	0.8606	0.8548	-0.5940	”
7	3000	0.7414	0.8599	0.8663	-2.1244	”

The good SVM model is the one with the parameters of rbf and 3000 c value (hyperparameter) as it has higher rscore compartively

3. Decision Tree:

S.No	Criterion	Max_features	Splitter	R_Score
1	squared_error	sqrt	best	0.7332
2	squared_error	sqrt	random	0.6785
3	squared_error	log2	best	0.7558
4	squared_error	log2	random	0.6349
5	squared_error	None	best	0.6963
6	squared_error	None	random	0.7400
7	friedman_mse	sqrt	best	0.7282
8	friedman_mse	sqrt	random	0.6319

9	friedman_mse	log2	best	0.7345
10	friedman_mse	log2	random	0.6517
11	friedman_mse	None	best	0.7083
12	friedman_mse	None	random	0.7159
13	absolute_error	sqrt	best	0.6566
14	absolute_error	sqrt	random	0.6847
15	absolute_error	log2	best	0.7253
16	absolute_error	log2	random	0.6174
17	absolute_error	None	best	0.6723
18	absolute_error	None	random	0.7205
19	poisson	sqrt	best	0.7491
20	poisson	sqrt	random	0.7257
21	poisson	log2	best	0.5865
22	poisson	log2	random	0.5983
23	poisson	None	best	0.7049
24	poisson	None	random	0.7557

The optimal decision tree model is configured with the squared_error criterion, max_features=log2, and splitter=best, since it obtained the highest R^2 score among the evaluated models.

4.Randomforest Rscore

S.No	Criterion	Max_features	R_Score
1	squared_error	sqrt	0.8690
2	squared_error	log2	0.8702
3	squared_error	None	0.8542
4	friedman_mse	sqrt	0.8702
5	friedman_mse	log2	0.8685
6	friedman_mse	None	0.8550
7	absolute_error	sqrt	0.8712
8	absolute_error	log2	0.8715
9	absolute_error	None	0.8518
10	poisson	sqrt	0.8703
11	poisson	log2	0.8701
12	poisson	None	0.8565

The optimal random forestmodel is configured with the absolute_error criterion and max_features= log2, since it obtained the highest R^2 score among the evaluated models.

The comparatively deployable model for this dataset is SVM model with the parameters of rbf kernel and 3000 c value (hyperparameter) as it has higher rscore comparatively.